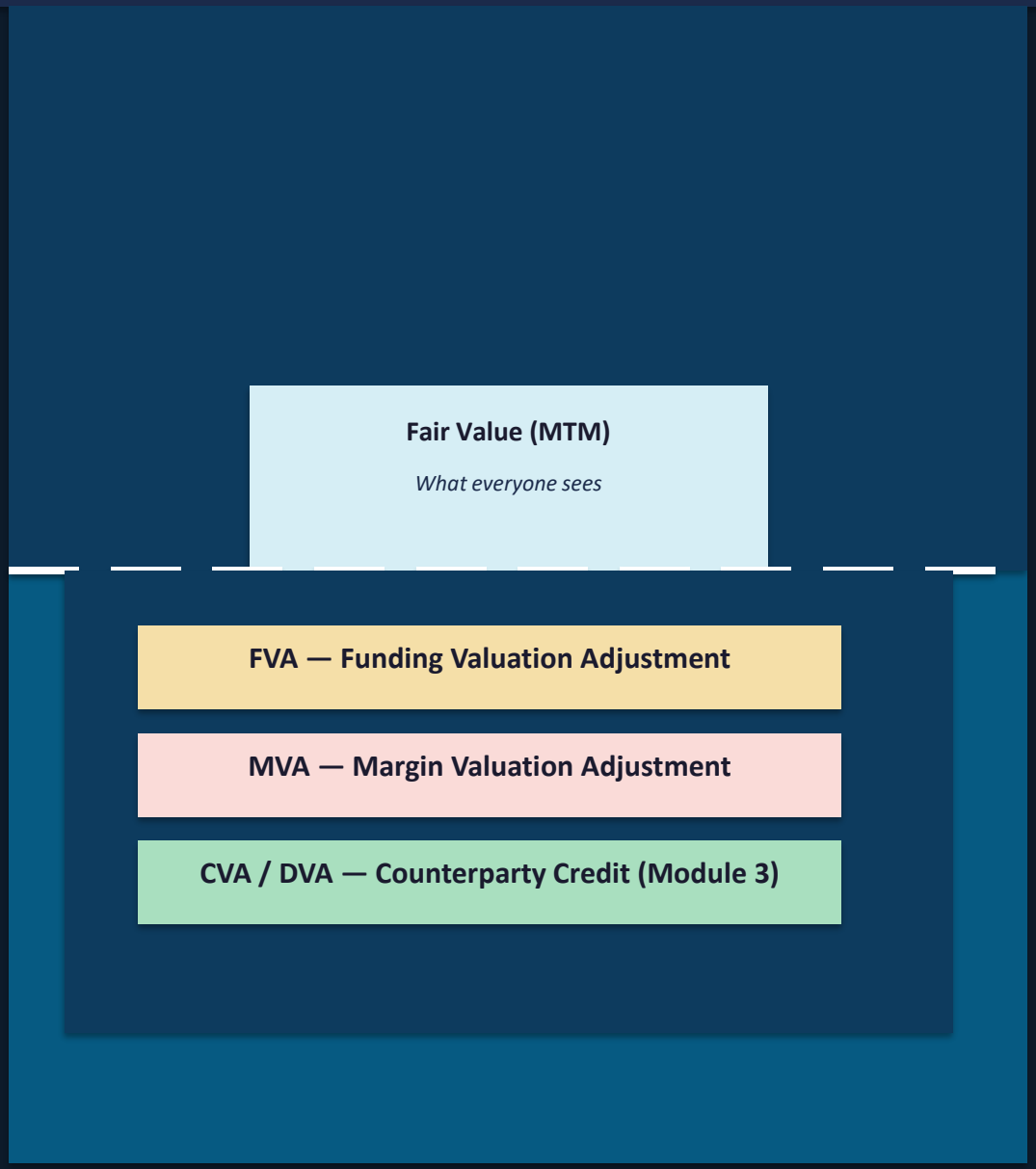


MODULE 4

# Funding Valuation Adjustment (FVA) & Margin Valuation Adjustment (MVA)

*The Hidden Tax on Every Derivative You Trade*

Sarthaka Nanda | with Anthropic Claude
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# The Iceberg Beneath Every Derivative

*Slides 1–3 of 4: What FVA & MVA are and why they matter*

## The Analogy

You lend a friend £100 — but you borrowed that £100 from your bank at 4%.

Even if your friend pays you back perfectly, you still paid the funding cost.

FVA is exactly that: the cost a dealer bank pays to fund derivative positions when the market doesn't automatically return collateral.

## MVA: The Pension Fund Problem

Post-2008 regulation (Uncleared Margin Rules / UMR) requires bilateral derivative users to post Initial Margin — cash locked with a custodian. It earns OIS. Your hurdle rate is higher. That drag over the life of the trade is MVA.

## Why Textbook Pricing Fails

### Textbook World

- Funding at risk-free rate
- Unlimited borrowing
- Collateral symmetry
- LIBOR ≈ OIS

### Real World (Post-2008)

- OIS + credit spread (30–150bps)
- Constrained funding capacity
- Asymmetric CSA terms
- Multi-curve divergence

The LIBOR-OIS spread hit 360bps in Oct 2008 — 35x its pre-crisis level. Banks pricing without FVA were booking hidden losses.

**JP Morgan booked a $1.5 billion FVA charge in Jan 2014.**
**The hidden tax finally became visible.**


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# FVA & MVA: Twin Costs, Different Origins

*Both price the cost of capital tied up in a derivative — from different sources*

### FVA — Funding Valuation Adjustment

- Arises from: uncollateralised derivative exposures
- Driver: dealer funding spread above OIS (30–150 bps)
- Applies to: bilateral OTC trades without full CSA
- Direction: FCA (cost) when ITM; FBA (benefit) when OTM
- Catalyst: 2008 multi-curve divergence & funding stress
- Key debate: Hull-White (2012) — 'FVA is a mistake'

### MVA — Margin Valuation Adjustment

- Arises from: mandatory Initial Margin under Uncleared Margin Rules (UMR)
- Driver: opportunity cost of segregated IM (OIS vs hurdle)
- Applies to: bilateral & cleared trades (post Phase 6 2022)
- Direction: always negative — IM posting is always a cost
- Catalyst: BCBS-IOSCO Uncleared Margin Rules 2016–2022 phase-in
- Standard: ISDA SIMM v2.8+2506 (EUR IR delta: 70–80 bps range)

$$FVA = -\int s_f(t) \cdot E[V+(t)] \cdot B(0,t)\, dt + \int s_f(t) \cdot E[V-(t)] \cdot B(0,t)\, dt$$

*s_f = funding spread; V+ = positive exposure (FCA cost); V- = negative exposure (FBA benefit)*

$$MVA = -\int s_m(t) \cdot E[IM(t)] \cdot B(0,t)\, dt$$

*s_m = funding cost minus IM remuneration rate; E[IM(t)] = expected SIMM IM at time t*

*Both are path-dependent integrals over expected future exposures — requiring Monte Carlo simulation of risk factors over the full trade lifetime.*


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# The Number That Changes Everything

*Why XVA desks exist: the all-in cost of a 'simple' swap*

## A EUR 100M 5-Year IRS

*(uncollateralised, bilateral UMR — Uncleared Margin Rules)*

| | |
|---|---|
| **CVA: −EUR 85,000** | Counterparty default risk (BBB, 100bps CDS) |
| **DVA: +EUR 32,000** | Own-default benefit (A-rated dealer, 50bps CDS) |
| **MVA: −EUR 103,000** | UMR bilateral IM, 80bps funding spread, SIMM |
| **FVA*: ~EUR 0** | At-the-money at inception; material after rate moves |
| **Total XVA: ≈ −EUR 156,000** | |

## What This Means

- A trade priced at-the-money on a risk-neutral basis carries EUR 156K in all-in XVA charges.
- Ignoring FVA & MVA means the trading desk books profits that treasury simultaneously pays out.
- After a 50bp adverse rate move, FVA alone generates ~EUR 480K additional cost.
- For a mid-size fund entering Phase 6 Uncleared Margin Rules (UMR) in 2022, MVA on a EUR 5B bilateral book = EUR 15–40M p.a.
- The full paper derives every component from first principles with worked numerical examples.

**Read the full paper → link in comments**


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# The Hull-White Controversy: Should FVA Exist?

*Risk Magazine, July 2012 — The most debated paper in modern derivatives*

## Hull & White (2012): FVA is a Mistake

- FVA is already captured in DVA (own-default benefit)
- Modigliani-Miller: funding cost is irrelevant to firm value
- Charging FVA destroys positive NPV trades
- Creates client-discriminatory, non-arbitrage-free pricing
- Risk-neutral framework is internally complete
- FVA is a shareholder transfer, not a real economic cost

## Industry Response: FVA IS Real

- M-M assumes perfect markets — violated post-2008
- DVA cannot be monetised without the bank defaulting
- Treasury charges desks real funding costs every day
- Banks with high spreads genuinely cannot compete
- Ignoring FVA creates systematic hidden P&L drag
- Andersen-Duffie-Song (2019): FVA belongs in pricing

**Resolution: JP Morgan booked $1.5B FVA loss in Jan 2014. Goldman, BofA, Morgan Stanley followed. The practitioners won.**


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# FVA: Formal Framework & Formula

*Piterbarg (2010) · Burgard-Kjaer (2011) · Green (2015)*

*The Core Insight: When funding isn't free, the risk-neutral discount rate must be replaced*

## Step 1: Piterbarg PDE (2010) — funding-adjusted pricing equation

$$\partial V/\partial t + \tfrac{1}{2}\sigma^2 S^2 \partial^2 V/\partial S^2 + r_f \cdot S \cdot \partial V/\partial S - r_c \cdot V = 0$$

*$r_f$ = dealer funding rate = OIS + $s_f$; $r_c$ = collateral rate (OIS for full CSA) → No CSA: discount at $r_f$, not OIS*

## Step 2: FVA = FCA + FBA — the two components (Burgard-Kjaer 2011)

| Component | Economic Meaning | Sign | Formula |
|---|---|---|---|
| FCA (Funding Cost Adj.) | Cost of funding positive exposure (trade is in-the-money → dealer borrows) | Negative (cost) | $-\int s_f(t) \cdot E[V+(t)] \cdot B(0,t)\, dt$ |
| FBA (Funding Benefit Adj.) | Benefit when out-of-the-money (counterparty effectively lends to dealer) | Positive (benefit) | $+\int s_f(t) \cdot E[V-(t)] \cdot B(0,t)\, dt$ |

$$\text{FVA (net)} = \int_0^T s_f(t) \cdot [\, E[V-(t)] - E[V+(t)] \,] \cdot B(0,t)\, dt$$

*At-the-money at inception: FCA ≈ FBA → Net FVA ≈ 0. After market moves: material asymmetry emerges.*


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# Worked Example: FVA on EUR 100M Uncollateralised IRS

*5-year EUR IRS · No CSA · Dealer funding spread = 80bps*

| Parameter | Value |
|---|---|
| Notional | EUR 100,000,000 |
| Maturity | 5 years |
| Fixed rate (dealer pays) | 3.50% p.a. |
| OIS rate | 3.20% |
| Dealer funding spread s_f | 80 bps |
| Rate volatility | 80 bps normal |

## Expected Exposure & FVA Calculation

| Year | EPE (EUR M) | B(0,t) | FCA (EUR) |
|---|---|---|---|
| 0.5 | 1.26 | 0.9841 | 49,582 |
| 1.0 | 1.77 | 0.9685 | 137,227 |
| 2.0 | 2.40 | 0.9380 | 180,096 |
| 3.0 | 2.71 | 0.9085 | 196,856 |
| 4.0 | 2.56 | 0.8800 | 179,917 |
| 5.0 | 1.41 | 0.8524 | 96,119 |
| TOTAL | | | ≈ EUR 1,537,653 |

**Key insight: At-the-money at inception → FCA = FBA → Net FVA ≈ 0**

**After 50bp adverse rate move:**
**FVA ≈ −EUR 480,000 (hidden funding cost)**

**Without FVA pricing: desk reports profit;**
**treasury reports funding shortfall. Same trade.**


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# MVA: Margin Valuation Adjustment — Framework

*Created by regulation: BCBS-IOSCO Uncleared Margin Rules (UMR) 2016–2022 · ISDA SIMM v2.8+2506*

**The Uncleared Margin Rules (UMR) Phase-In**

| Phase 1 2016 | Phase 2-4 2017-19 | Phase 5 2021 | Phase 6 2022 |
|---|---|---|---|
| Largest dealers | Major banks | Mid-size firms | USD 8B+ buy-side |

## ISDA SIMM: Sensitivity-Based IM Model

$$IM_class = \sqrt{\left( \Sigma_i \, \Sigma_j \, \rho_{ij} \cdot WS_i \cdot WS_j \right)}$$

*$WS_i$ = delta_i × RW_i (risk weight per SIMM v2.8+2506); $\rho_{ij}$ = tenor correlation matrix*

## MVA Formula

$$MVA = -\int_0^T s_m(t) \cdot E[IM(t)] \cdot B(0,t)\, dt$$

*s_m = dealer funding rate minus IM remuneration rate (typically OIS); always negative*

| Property | Detail |
|---|---|
| IM computation | ISDA SIMM (bilateral) or CCP SPAN/TIMS (cleared) |
| Remuneration | OIS/ESTR; some CSAs specify Fed Funds |
| Funding spread s_m | Typically 20–100 bps above IM remuneration |
| IM profile shape | Highest at inception; declines with remaining maturity |
| MVA sign | Always negative — posting IM is always a cost |
| Phase 6 impact | USD 1.3 trillion IM posted globally by end-2022 |

⚠ SIMM v2.8+2506 (Dec 2025): EUR IR delta RW confirmed within 70–80 bps range. Credit Qualifying weights increased (high-yield/non-rated). Commodity weights decreased. 75 bps used in worked examples remains valid. Refer to current ISDA SIMM publication for live production calibration.


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# Worked Example: MVA on Same EUR 100M IRS

*Bilateral Uncleared Margin Rules (UMR) · ISDA SIMM IM · s_m = 80bps · Linear IM amortisation*

### Step 1: SIMM IM at Inception

EUR 5Y IRS delta sensitivity (DV01 × 100bps scale):

DV01 ≈ EUR 45,000 per bp
Scale to SIMM: × 100 = EUR 4,500,000

SIMM Risk Weight (EUR 5Y bucket):
v2.8+2506: EUR IR delta ≈ 70–80 bps range
Example uses 75 bps (within confirmed range)

SIMM IM ≈ EUR 4,500,000 × 75bps
≈ EUR 3,375,000

**SIMM_IM ≈ Delta × RW_IR = EUR 3,375,000**

*Simplified single-bucket; actual SIMM aggregates across tenors with correlation offsets*

IM declines linearly to zero at maturity as residual risk falls with remaining tenor

### MVA Calculation (Semi-Annual)

| Year t | E[IM(t)] EUR M | s_m (bps) | MVA contrib (EUR) |
|---|---|---|---|
| 0.5 | 3.038 | 80 | 11,960 |
| 1.0 | 2.700 | 80 | 20,877 |
| 2.0 | 2.025 | 80 | 15,194 |
| 3.0 | 1.350 | 80 | 9,812 |
| 4.0 | 0.675 | 80 | 4,752 |
| 5.0 | 0.000 | 80 | 0 |
| TOTAL | | | ≈ EUR 102,660 |

**Total XVA = CVA (-85K) + DVA (+32K) + MVA (-103K) = -EUR 156K**

*Full XVA charge on a trade that appears flat on risk-neutral basis*


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# Regulatory Treatment: Uncleared Margin Rules (UMR), FRTB-CVA & SA-CCR

*How regulation creates, shapes, and complicates FVA & MVA*

| Regulation | Jurisdiction | FVA/MVA Impact |
|---|---|---|
| BCBS-IOSCO Uncleared Margin Rules — UMR (2015) | Global | Primary creator of MVA — mandatory bilateral IM requirement |
| EMIR Refit | EU | Phase 5/6 brings pension funds in scope; significant MVA increase |
| Dodd-Frank / SEC Rule (2022) | US | Extended scope to equity derivatives; new MVA populations |
| SA-CCR (Basel III) | Global | Higher EAD → increased cleared IM requirements → higher MVA |
| FRTB-CVA Framework | Global | FVA hedges NOT recognised as CVA hedges → regulatory/economic basis |
| CRR III (EU, 2025) | EU | Output floor increases RWA; indirectly raises funding costs & FVA |
| IFRS 13 / ASC 820 | Global | 'Exit price' fair value effectively mandates FVA inclusion |

**The Clearing Decision: Total XVA = CVA + FVA + MVA + KVA must be computed for bilateral vs cleared**

### Bilateral OTC

- CVA/DVA applies
- FVA applies (partial/no CSA)
- MVA: SIMM IM (higher, segregated)
- Netting within set only

### Cleared (CCP)

- CVA = 0 (CCP guarantee)
- FVA near-zero (daily VM)
- MVA: CCP model (often lower, nettable)
- Portfolio netting across members

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# Crisis Case Studies: When FVA & MVA Became Real

*Lehman 2008 · JP Morgan 2014 · Uncleared Margin Rules Phase 6 (2022)*

### Lehman Brothers
### September 2008

- 900K+ contracts, >$35T notional — no robust CSAs
- Funding spread hit 300bps+ in Aug 2008 (FCA squeeze)
- Repo markets froze: derivatives funding evaporated overnight
- Close-out losses ran for years → FVA should have been reserved from Day 1

### JP Morgan
### $1.5B FVA Charge 2014

- First bank to publicly recognise and book FVA loss
- Goldman, BofA, Morgan Stanley followed within months
- Industry had been underpricing uncollateralised trades for years
- Hull-White debate effectively ended at practitioner level

### Uncleared Margin Rules
### Phase 6 · Sep 2022

- $1.3 trillion in bilateral IM posted globally by end-2022
- Mid-size pension funds, hedge funds hit for first time
- EUR 5B fund: MVA cost increase of EUR 15–40M p.a.
- Drove major clearing decisions and trade compression wave


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# Advanced Topics: FVA Asymmetry, ColVA & XVA Optimisation

*Beyond the basics: the frontier of XVA practice*

### FVA Wrong-Way Risk

When funding spread widens at the same time exposure increases, true FVA is higher than independent calculation suggests. Requires joint simulation of market risk factors and dealer credit spread — computationally intensive but increasingly required.

### Bilateral vs Unilateral FVA

Unilateral FVA: only the dealer's costs → asymmetric pricing. Bilateral FVA: both parties' costs averaged → symmetric but requires counterparty funding info. Burgard-Kjaer semi-replication avoids DVA/FVA double-count by construction.

### ColVA — Preview of Module 6

Collateral Valuation Adjustment: the value of choosing the cheapest-to-deliver asset from the CSA-eligible schedule. A bank with a large liquid bond portfolio faces lower effective s_m than one posting cash IM. Full treatment in Module 6.

**XVA Optimisation: Trade compression · Netting set routing · Clearing decisions · CSA renegotiation · Collateral transformation**


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# Credits & Methodology

## Author

Sarthaka Nanda

## The 20-60-20 Principle

**20%** — **Human prompting & instruction**

*Topic selection, structural design, worked example parameters, strategic framing*

**60%** — **AI generation**

*Technical content, formula derivation, tables, regulatory mapping, case studies*

**20%** — **Human review & editing**

*Mandatory review for accuracy, consistency, and professional standards before publication*

***AI-generated content without mandatory human review is not professional-grade content — it is a liability.***

## The Series

- ✓ Module 1: Basel Accord Evolution
- ✓ Module 2: FRTB
- ✓ Module 3: CVA, DVA & CCR
- ★ Module 4: FVA & MVA ← YOU ARE HERE
- ○ Module 5: KVA & Capital Costs (forthcoming)
- ○ Module 6: ISDA, Netting & ColVA (forthcoming)

# Glossary of Key Terms

*Acronyms and technical terms used throughout this module*

| | |
|---|---|
| **BA-CVA** | Basic CVA — simplified regulatory capital approach for CVA risk under FRTB |
| **BCBS** | Basel Committee on Banking Supervision — sets global banking regulatory standards |
| **CCR** | Counterparty Credit Risk — risk of loss if a derivative counterparty defaults |
| **CCP** | Central Counterparty — clearing house that interposes itself between buyer and seller |
| **CET1** | Common Equity Tier 1 — highest quality regulatory capital under Basel III |
| **ColVA** | Collateral Valuation Adjustment — value of cheapest-to-deliver collateral optionality |
| **CSA** | Credit Support Annex — legal agreement governing collateral posting between parties |
| **CVA** | Credit Valuation Adjustment — price of counterparty default risk in a derivative |
| **DVA** | Debit Valuation Adjustment — benefit from own-default risk embedded in a liability |
| **EAD** | Exposure At Default — estimated exposure when a counterparty defaults |
| **EMIR** | European Market Infrastructure Regulation — EU derivatives clearing & reporting rules |
| **ENE** | Expected Negative Exposure — average of negative (out-of-the-money) exposure values |
| **EPE** | Expected Positive Exposure — average of positive (in-the-money) exposure values |
| **FBA** | Funding Benefit Adjustment — benefit component of FVA when trade is out-of-the-money |
| **FCA** | Funding Cost Adjustment — cost component of FVA when trade is in-the-money |
| **FRTB** | Fundamental Review of the Trading Book — Basel IV market risk capital framework |
| **FVA** | Funding Valuation Adjustment — cost of funding uncollateralised derivative exposures |
| **IM** | Initial Margin — upfront collateral buffer posted against future potential exposure |
| **ISDA** | International Swaps and Derivatives Association — derivatives trade body & standards |
| **ISDA SIMM** | ISDA Standard Initial Margin Model — sensitivity-based IM model for bilateral trades |
| **KVA** | Capital Valuation Adjustment — cost of regulatory capital consumed by a trade |
| **LIBOR** | London Interbank Offered Rate — legacy benchmark rate, largely replaced by RFRs |
| **MVA** | Margin Valuation Adjustment — opportunity cost of posting initial margin over trade life |
| **OIS** | Overnight Index Swap — swap referencing overnight rate; standard discount rate post-2008 |
| **RFR** | Risk-Free Rate — overnight benchmarks replacing LIBOR (e.g. ESTR, SOFR, SONIA) |
| **RWA** | Risk-Weighted Assets — basis for calculating minimum regulatory capital requirements |
| **SA-CCR** | Standardised Approach for CCR — Basel III method for computing derivative EAD |
| **SA-CVA** | Standardised CVA — sensitivity-based regulatory capital approach for CVA under FRTB |
| **SIMM** | Standard Initial Margin Model — see ISDA SIMM above |
| **UMR** | Uncleared Margin Rules — BCBS-IOSCO rules requiring IM on non-cleared derivatives |
| **VM** | Variation Margin — daily mark-to-market collateral exchanged to settle P&L changes |
| **XVA** | Collective term for all Valuation Adjustments: CVA, DVA, FVA, MVA, KVA, ColVA |

All formulas use continuous-time notation unless marked discrete. $s^o$ = funding spread; $s_m$ = MVA spread; B(0,t) = OIS discount factor; $\Delta t$ = time step


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